

Conjugated Systems & Aromaticity

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1 Aromaticity & Aromatic Substitution

1.1 Aromatic Compounds

Core Concept 1.1: Aromatic Compounds

The name "aromatic" originates from these compounds' characteristic scent; however, the term now denotes a strictly structural definition, as covered in Core Concept 1.5.

While aromaticity is not limited to the base benzene structure, these notes will primarily refer to benzene-derived compounds such as the following.

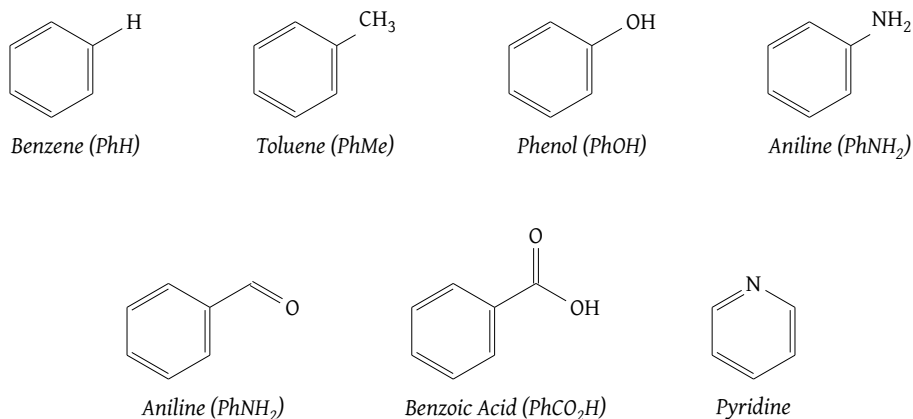


Figure 1: Phenyl ring-based aromatic compound examples; pyridine represents an aromatic compound with a non-CH atom as part of its ring

Core Concept 1.2: Aromatic Compound Stereochemistry

Additional substituent positions on the root aromatic compounds shown above can be specified using the ortho, meta, and para keywords.

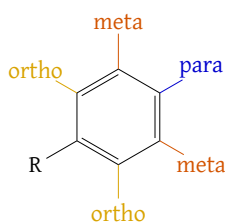


Figure 2: Phenyl ring substituent-positioning keywords

The compound naming scheme follows the structure *relation-substituent-root*. Consider the following examples.

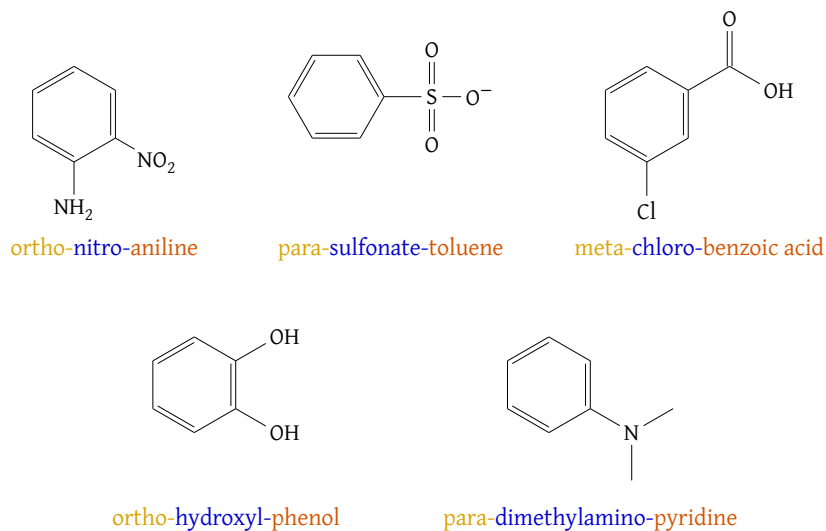


Figure 3: Phenyl ring-based aromatic compound nomenclature

Core Concept 1.3: Reactivity of Aromatic Compounds

Aromatic compounds are considerably less reactive than would otherwise be predicted. For instance, while a simple cyclohexene reacts readily with mCPBA to form an epoxide, no reaction occurs when benzene is mixed with three equivalents of mCPBA.

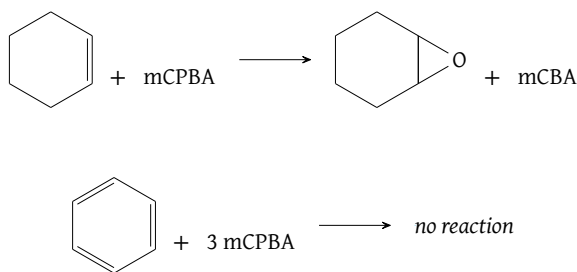


Figure 4: Reactions of cyclohexene and benzene with mCPBA

Similarly, a substantial portion of the predicted energy release during the hydrogenation of benzene is offset by aromatic stabilization.

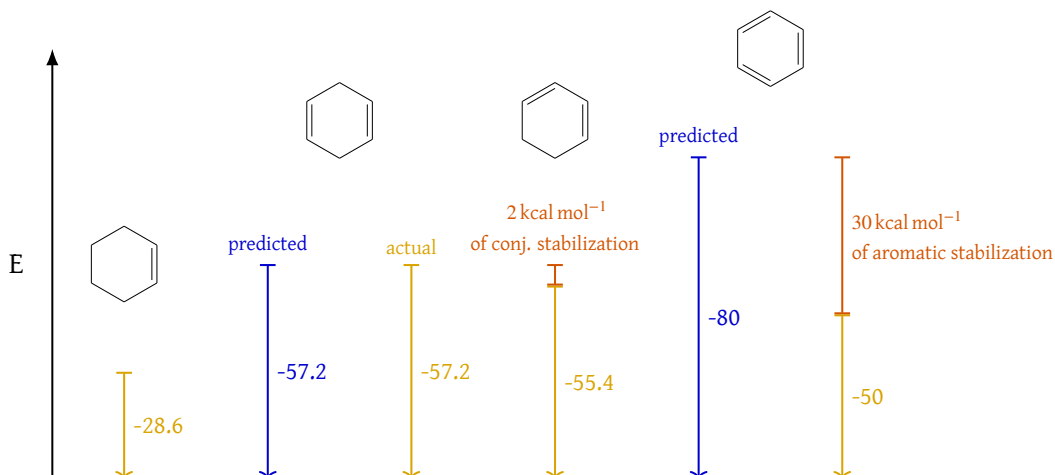


Figure 5: Energy diagram of predicted versus actual heats of hydrogenation for cyclohexene derivatives; aromaticity confers a substantial increase in stability relative to the summed values of the corresponding isolated double-bond hydrogenations, exceeding even the stabilization observed in nonaromatic but conjugated cyclohexene systems

To explain this phenomenon, consider a top-down view of the benzene π molecular orbital diagram.

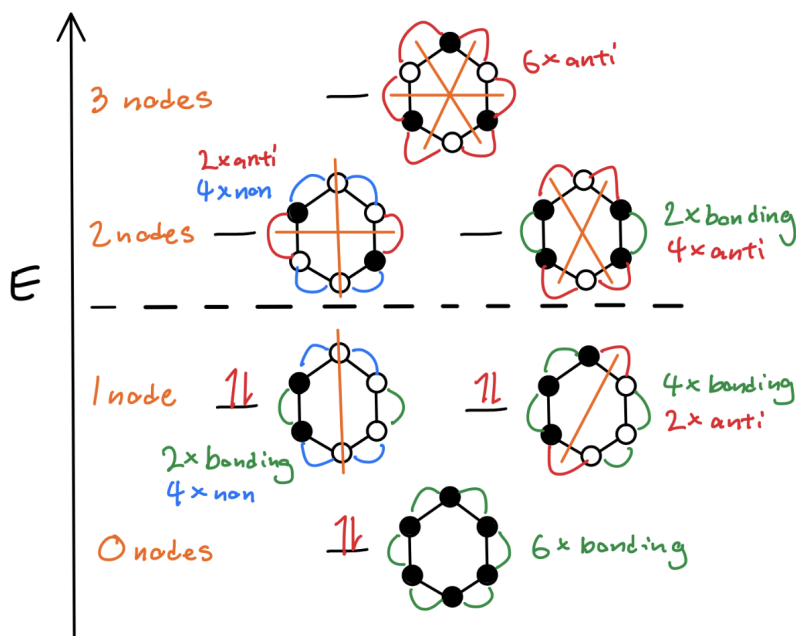


Figure 6: Energy diagram of the benzene π molecular orbitals (top-down view)

As shown in Figure 6, the π electrons occupy exclusively bonding molecular orbitals, rendering the system exceptionally stable.

Core Concept 1.4: Aromatic Compound Molecular Diagrams

The **Frost Diagram** is a method for rapidly predicting π molecular orbital energy levels of a cyclic conjugated system, following the steps below.

- (1) The compound must possess a cyclic array of p orbitals
- (2) Draw the ring structure with one vertex pointed downwards
- (3) Draw a circle around the compound
- (4) Intersections represent energy levels
- (5) The middle of the circle represents the nonbonding energy level
- (6) Add e^- s

Below are the corresponding orbital diagrams for a cyclopentenyl anion and a cycloheptenyl cation; the benzene diagram has already been shown above.

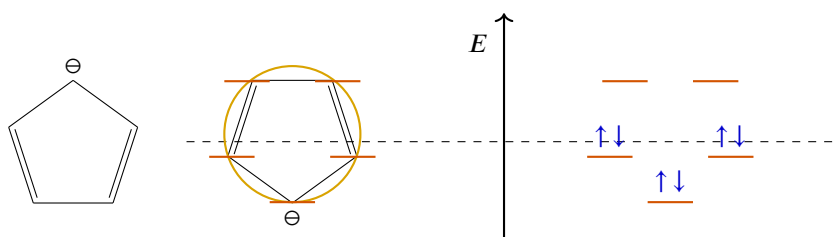
(a) Cyclopentene Anion

Figure 7: Frost diagram for the cyclopentenyl anion, showing three filled bonding/nonbonding orbitals and two unoccupied, degenerate antibonding orbitals

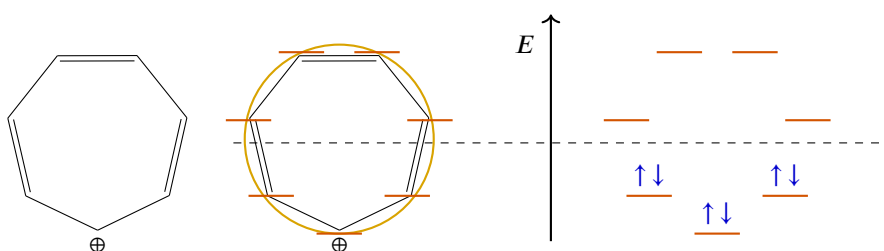
(b) Cycloheptene Cation

Figure 8: Frost diagram for the cycloheptenyl cation, showing three filled bonding/nonbonding orbitals and four unoccupied, degenerate antibonding orbitals

Core Concept 1.5: Aromaticity Rules

In order for a compound to be considered aromatic, it must satisfy a strict set of rules.

- (1) The compound must be a cyclic array of p orbitals
- (2) The e^- count of the cyclic array must fulfill the equation of $4n + 2$, where n is an integer (0, 1, 2, 3, 4, ...).

*An e^- count that instead fulfills the equation $4n$, where n is an integer, results in a compound that is far more reactive than predicted and is termed **anti-aromatic**.*

If the above criteria are not met, the compound is considered **nonaromatic**, as in the cyclohexene shown below.

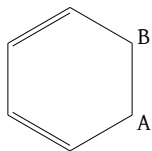


Figure 9: Nonaromatic cyclohexene; carbons A and B are sp^3 , which prevents an uninterrupted cyclic array of p orbitals

As practice, consider the following two examples.

(a) **Pentene Anion**

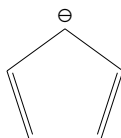


Figure 10: Aromatic pentene anion

This structure exhibits a cyclic array of p orbitals with an electron count of 6, satisfying the $4n + 2$ expression. This compound is therefore **considered aromatic**.

(b) **Pentene Cation**

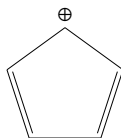


Figure 11: Anti-aromatic pentene cation

This compound likewise exhibits a cyclic array of p orbitals, but its electron count of 4 satisfies $4n$ rather than $4n + 2$. This compound is therefore considered **anti-aromatic**.

Notice that the corresponding molecular orbital diagram results in two unpaired electrons, accounting for its heightened reactivity.

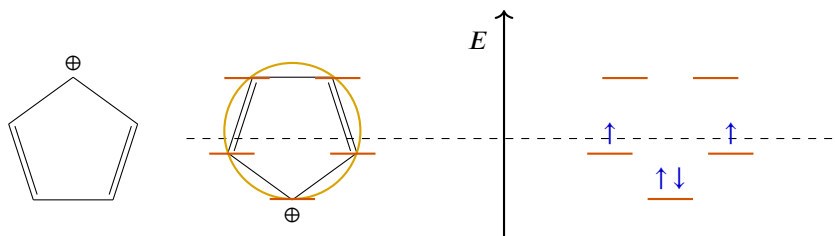


Figure 12: Frost diagram for the cyclopentenyl cation

Core Concept 1.6: Aromatic Contributors

An **aromatic contributor** refers to a resonance structure of a molecule that possesses a fully conjugated, cyclic, planar π system satisfying the $4n + 2$ π -electron count, even when the dominant or lowest-energy Lewis

structure does not immediately appear aromatic. Consider the following molecule as an example.

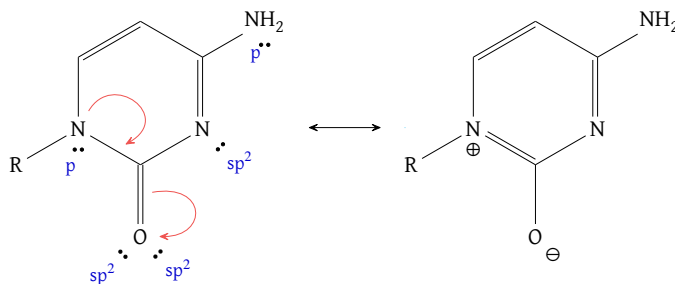


Figure 13: Aromatic contributor; a simple rule of thumb for identifying a resonance contributor is to attempt to achieve alternating π bonds while ensuring that a circle drawn around the ring never crosses a π bond

Notice that although the base resonance structure is nonaromatic by definition, since its conjugated π orbital system is not cyclic, an alternative structure accessible through resonance is aromatic, thereby qualifying the molecule as possessing an aromatic contributor.

1.2 Electrophilic Aromatic Substitution

Core Concept 1.7: Electrophilic Aromatic Substitution

Electrophilic aromatic substitution (EAS) describes a class of reactions in which an aromatic ring functions as the nucleophilic species. The general mechanism underlying this reaction type is outlined below.

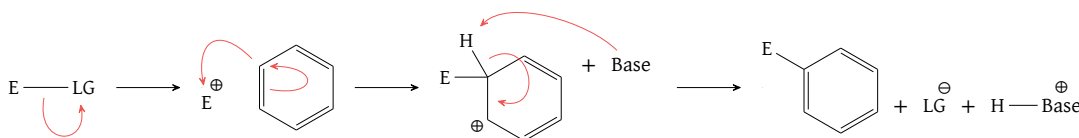


Figure 14: General mechanistic outline of electrophilic aromatic substitution reactions

1.2.1 Reaction Types

Core Concept 1.8: Halogenation

EAS halogenation results in the attachment of a single halogen atom to an aromatic ring, with the corresponding hydrohalic acid generated as a byproduct. This mechanism proceeds via activation by a **Lewis acid** (defined as any chemical species capable of accepting an electron pair), as illustrated below.

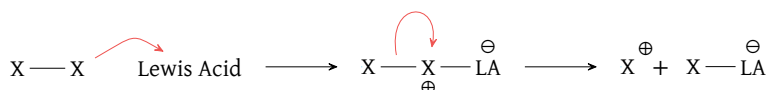


Figure 15: Generation of a halogen cation electrophile via Lewis acid activation; Lewis acids function as electron-pair acceptors and include species such as FeX_3 , BX_3 , and AlX_3

The chlorination and bromination of benzene, presented below, illustrate this general mechanism.

(a) Chlorination of Benzene

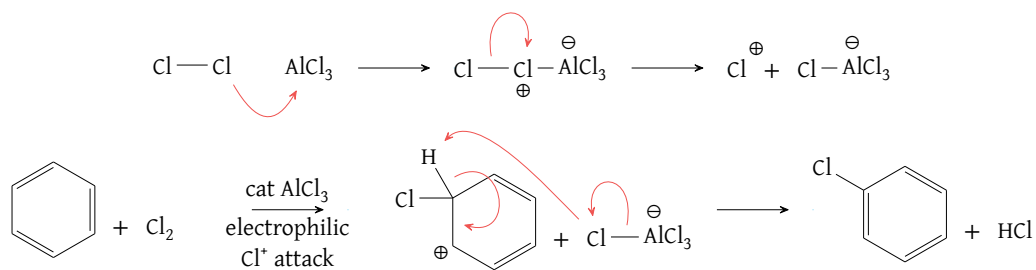


Figure 16: Chlorination of benzene

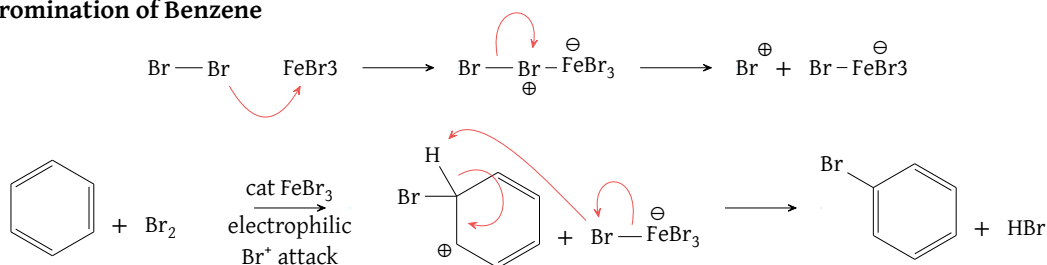
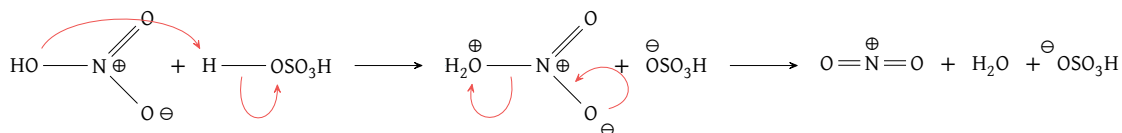
(b) Bromination of Benzene

Figure 17: Bromination of benzene

Core Concept 1.9: Nitration

Nitration involves the attachment of a nitro group (NO_2) to an aromatic ring and serves primarily as a synthetic intermediate step in practice. Consider first the generation of the NO_2^+ electrophile, formed upon exposure of nitric acid to sulfuric acid.

Figure 18: Generation of the NO_2^+ electrophile

With this electrophile established, the nitration of benzene proceeds according to the general mechanism outlined in Core Concept 1.7.

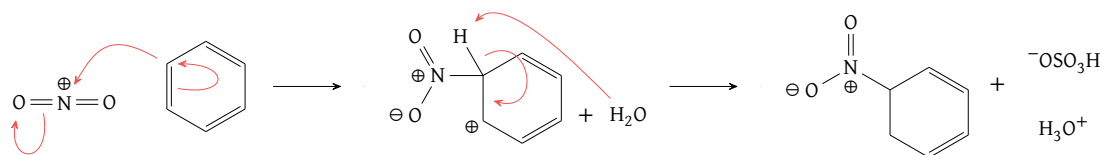
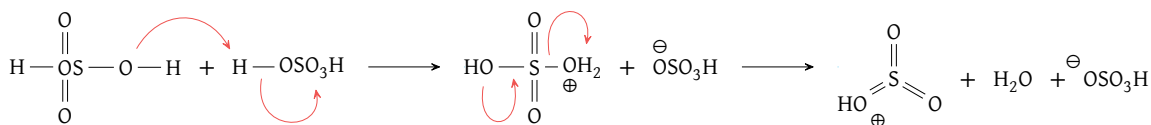


Figure 19: Nitration of benzene

Core Concept 1.10: Sulfonation

Sulfonation entails the attachment of a sulfonic acid group (SO_3H) to an aromatic ring. The initial step involves a reaction between two molecules of H_2SO_4 , arising from the use of **fuming H_2SO_4** as one of the reagents.

Figure 20: Formation of fuming H_2SO_4

The reaction then proceeds according to the same general EAS mechanism described in Core Concept 1.7.

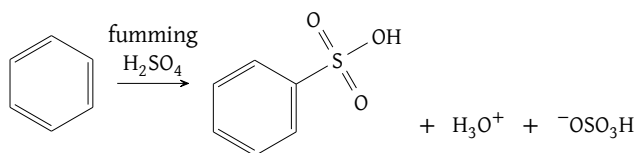
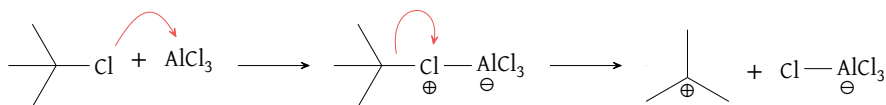


Figure 21: Sulfonation of benzene

Core Concept 1.11: Alkylation

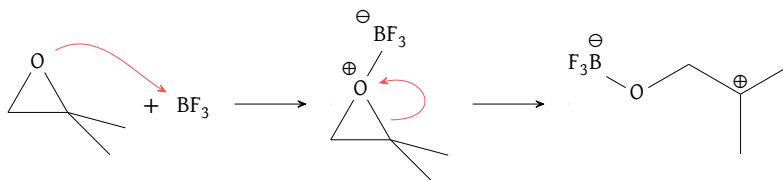
Alkylation refers to the addition of an alkyl group to an aromatic structure. Before an alkane can act as an electrophile, it must first undergo conversion to a carbocation intermediate, as shown in the reactions below.

(a) Generation of an Electrophile via a Lewis Acid

Figure 22: Mechanism for the formation of a tert-butyl carbocation from tert-butyl chloride using aluminum chloride (AlCl_3) as a Lewis acid catalyst

This transformation parallels the initial step of the chlorination pathway presented in Core Concept 1.8.

(b) Lewis Acid-Catalyzed Epoxide Ring Opening

Figure 23: Boron trifluoride (BF_3) coordination to an epoxide, promoting regioselective ring cleavage to generate the more thermodynamically stable tertiary carbocation

(c) Acid-Catalyzed Carbocation Formation from an Alcohol

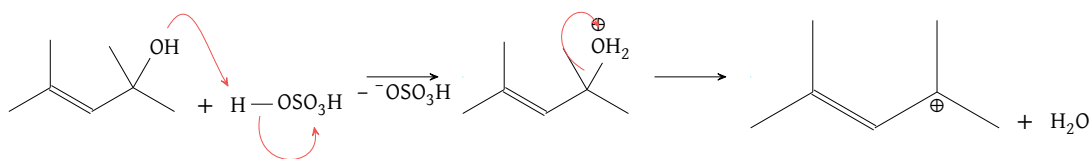
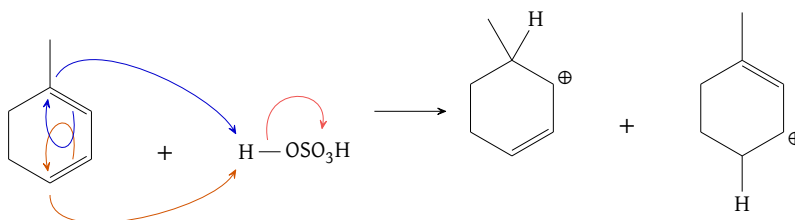


Figure 24: Protonation of a tertiary alcohol by sulfuric acid, followed by the expulsion of water to form a tertiary carbocation intermediate

(d) Protonation of an Aromatic Ring

Figure 25: Electrophilic attack on toluene by H^+ ; the first hydrogenation step follows either the blue or orange e^- movement

In certain cases, carbocation rearrangement occurs when the initially formed carbocation is unstable.

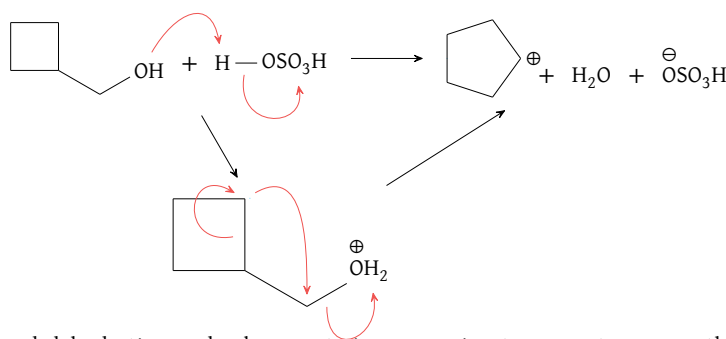


Figure 26: Acid-catalyzed dehydration and subsequent ring expansion to generate a more thermodynamically stable cyclopentyl carbocation

These rearranged carbocations subsequently undergo EAS with aromatic structures.

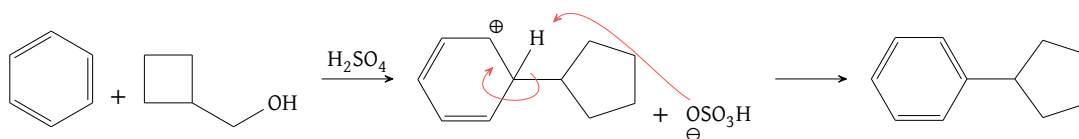


Figure 27: Electrophilic aromatic substitution of benzene by a rearranged cyclopentyl carbocation

In the case above, illustrating rearrangement from a cyclobutyl to a cyclopentyl carbocation, such rearrangement can be avoided through a stepwise pathway involving **acylation** (Core Concept 1.12) followed by reduction with Zn(Hg) and HCl .

(a) Acylation

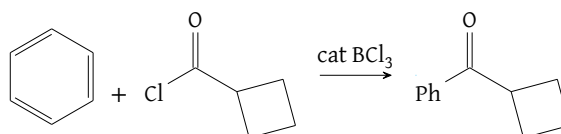


Figure 28: Acylation

(b) Reduction

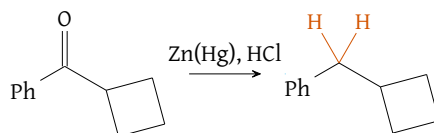


Figure 29: Reduction of ketone with Zn(Hg) and HCl

Core Concept 1.12: Acylation

Acylation refers to the addition of an acyl halide to an aromatic structure. The acyl halide must first be rendered electrophilic through activation by a Lewis acid, as shown in the general reaction outline below.

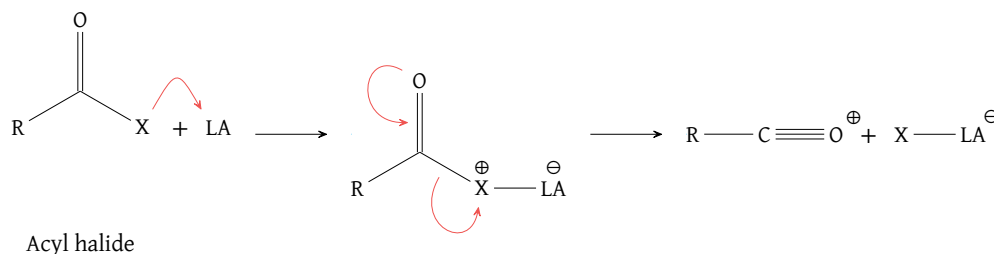


Figure 30: Conversion of an acyl halide to an electrophile

The resulting RCO^+ species can then undergo EAS with an aromatic ring, as demonstrated below using representative compounds.

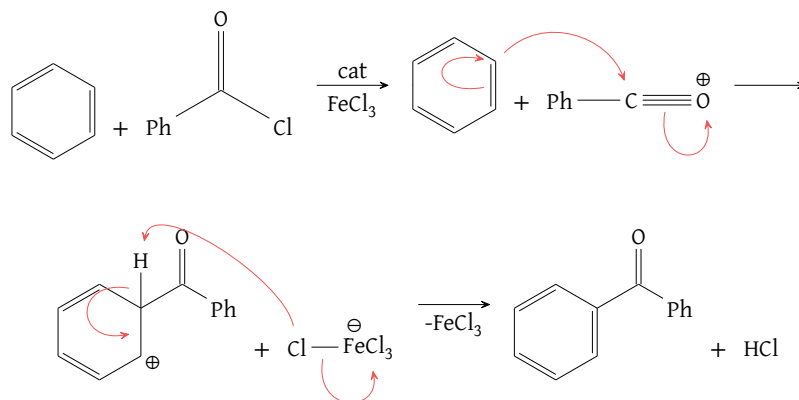


Figure 31: Acylation of benzene

1.2.2 Directing Groups**Core Concept 1.13: Directing Groups**

So far, we have covered simply the EAS reaction types in Subsubsection 1.2.1 namely using benzene as our sole relevant aromatic structure. However, in scenarios where the aromatic ring is asymmetrical, chemists need a method to determine which carbon they want the EAS to occur on. Now we introduce **directing groups**, functional groups already attached to a benzene ring that dictate where (O, M, P) a new incoming electrophile will land.

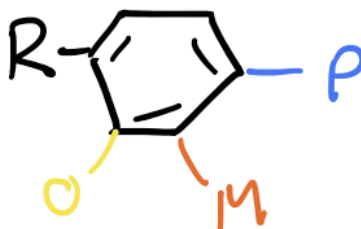


Figure 32



Figure 33: caption

A summary table is provided below, but visit each Core Concept for a more detailed look into each directing group type.

Directing Group Type	Examples	Description
(1) Inductive O / P Director	$R = \text{alkyl [Me, Et, } i\text{Pr, ...]}$	Alkyl groups donate electron density inductively, directing substitution to the ortho and para positions.
(2) Inductive M Director	$R = \text{carbon w/ EN atoms [CCl}_3, \text{CF}_3, \text{...}]}$	Carbons bearing electronegative atoms withdraw electron density inductively, directing substitution to the meta position.
(3) Resonance O / P Director	i) R has lone pair adjacent to ring: [OMe, SR, NMe ₂] ii) $R = \text{carbon w/ } \pi \text{ system: [alkene, alkyne, benzene]}$	Substituents that donate electron density via resonance into the ring direct substitution to the ortho and para positions.
(4) Resonance M Director	[C=O, C≡N, NO ₂]	Substituents that withdraw electron density via resonance from the ring accept electron density and direct substitution to the meta position.

Table 1: Classification of electrophilic aromatic substitution directing groups

Core Concept 1.14: Inductive O/P Directors

Inductive O/P directors are substituents that direct any EAS to the O and P positions within an aromatic ring.

① Inductive O/P Directors

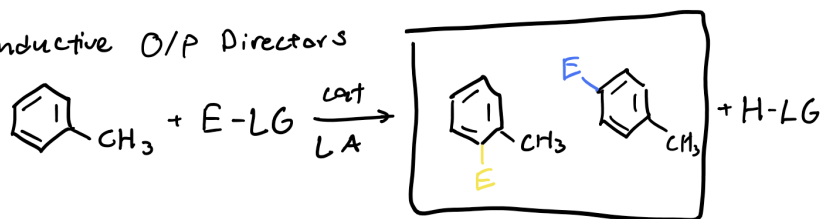


Figure 34

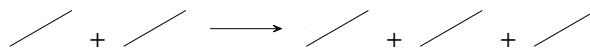


Figure 35: caption

The substituents are generally alkyls, and the mechanism behind why they favor O/P over M positioning is due to the general stability of the different resonance structures that occur in each positioning scenario.

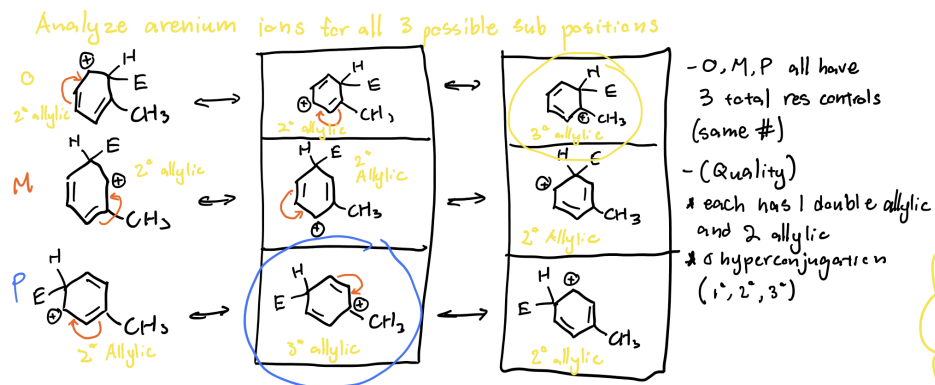


Figure 36

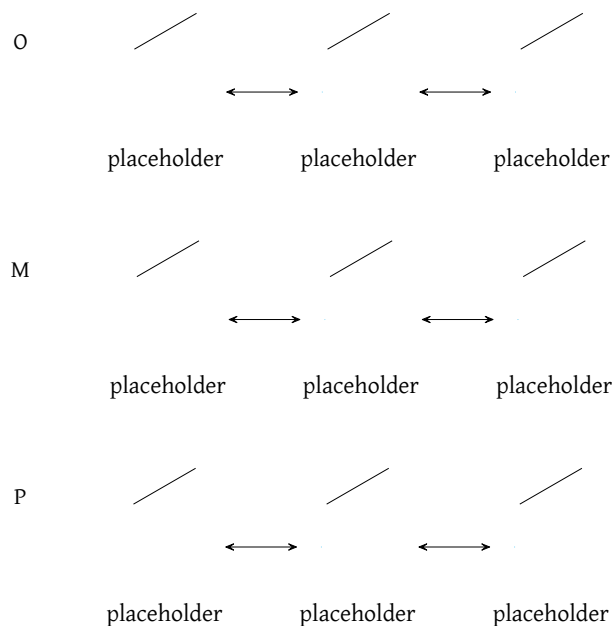


Figure 37: caption

As shown above, while nearly all of the different possible resonance structures place the transient carbocation charge on a 2° allylic position, O/P positioning allows for a 3° allylic carbocation position within 1 out of 3 resonance structures for each whereas M positioning does not. As such the energy levels of the resulting carbocation intermediaries follow the general ordering shown in the energy diagram below.

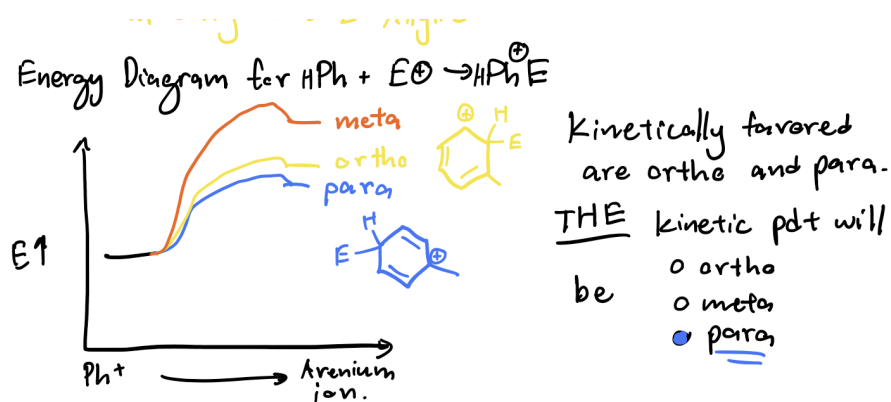


Figure 38

Core Concept 1.15: Inductive M Directors

Inductive M directors are substituents that direct any EAS to the M position within an aromatic ring. The substituents are carbons bearing highly electronegative atoms, making the carbon highly partially positive.

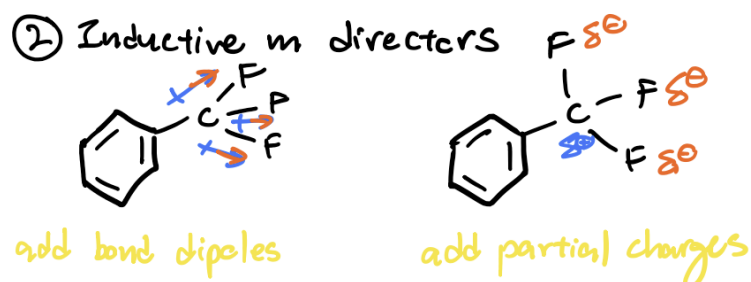


Figure 39

Figure 40: caption

The M position is favored because the O/P 3° allylic carbocation resonance structures have possess extra unstable repulsion between the carbocation and the partial positive carbon within the substituent. The M positioning, on the other hand, possess for a resonance structure that allows for maximal distance between the positive substituent carbon and the carbocation charge on teh opposite side fo the aromatic ring.

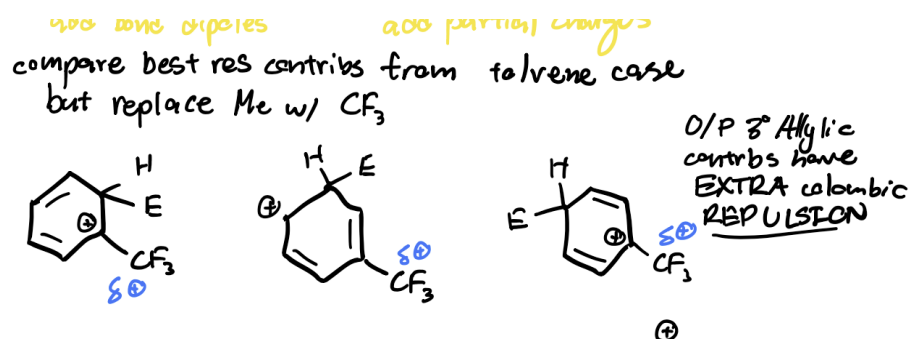


Figure 41

Figure 42: caption

Consequently, the resulting energy levels put O/P positioning energy above that of M positioning.

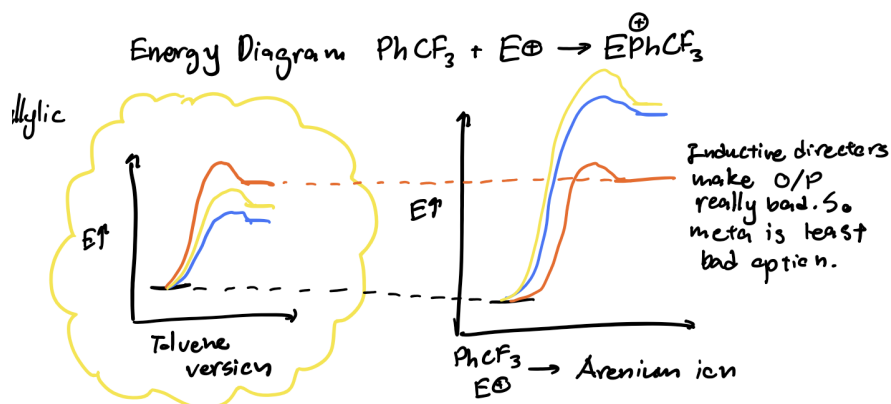


Figure 43

Core Concept 1.16: Resonance O/P Directors

Resonance O/P directors are substituents that direct any EAS to the O and P positions within an aromatic ring through donating e^- density. For instance, phenol possess resonance structures that place a carboanion negative charge on exclusively the O and P positions.

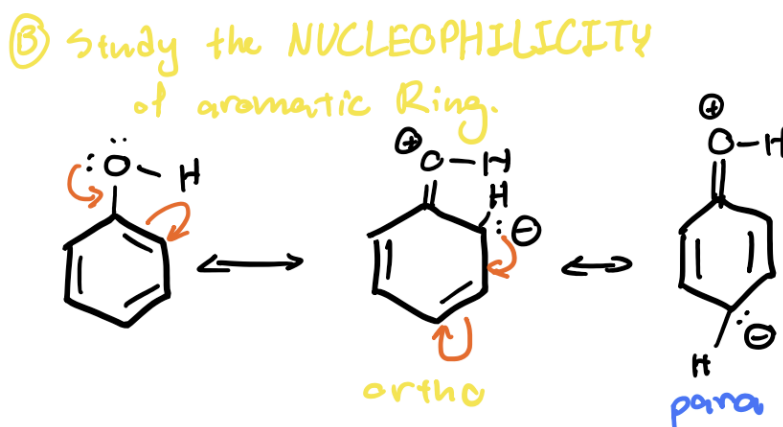


Figure 44

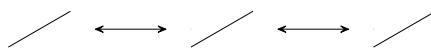


Figure 45: caption

These types of substituents specifically need to be lone pair adjacent to the aromatic ring. Another substituent type that allows for e^- donation is carbon with conjugated π systems. Take the following alkene for instance.

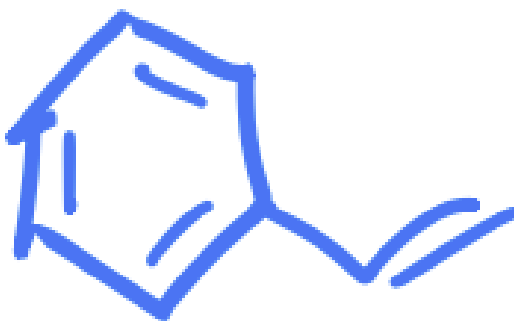


Figure 46: show the resonance possible



Figure 47: caption

Notice that the primary carbocation is able to form because it is allylic.

Here is a side note regarding the method by which inductive directors induce favorability of EAS positioning in comparison to resonance directors -> resonance directors affect the nucleophilicity capabilities of the relevant positions whereas the inductive versions manipulate favorability through transition state stability.

Core Concept 1.17: Resonance M Directors

Resonance M directors are substituents that direct any EAS to the M position within an aromatic ring through withdrawing e^- density. For instance, carbonyl substituents have the ability to withdraw e^- density in a manner that results in subsequent resonance structures to place a carbocation positive charge on the O/P positions, making these locations effectively less nucleophilic and the P position relatively more nucleophilic.

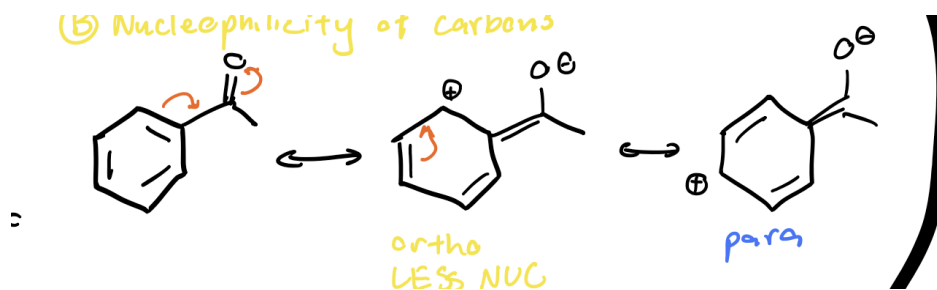


Figure 48

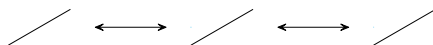


Figure 49: caption

As a result, when an electrophile approaches the transient aromatic ring, the nucleophilic attack occurs most favorably from the M position rather than the O/P position.

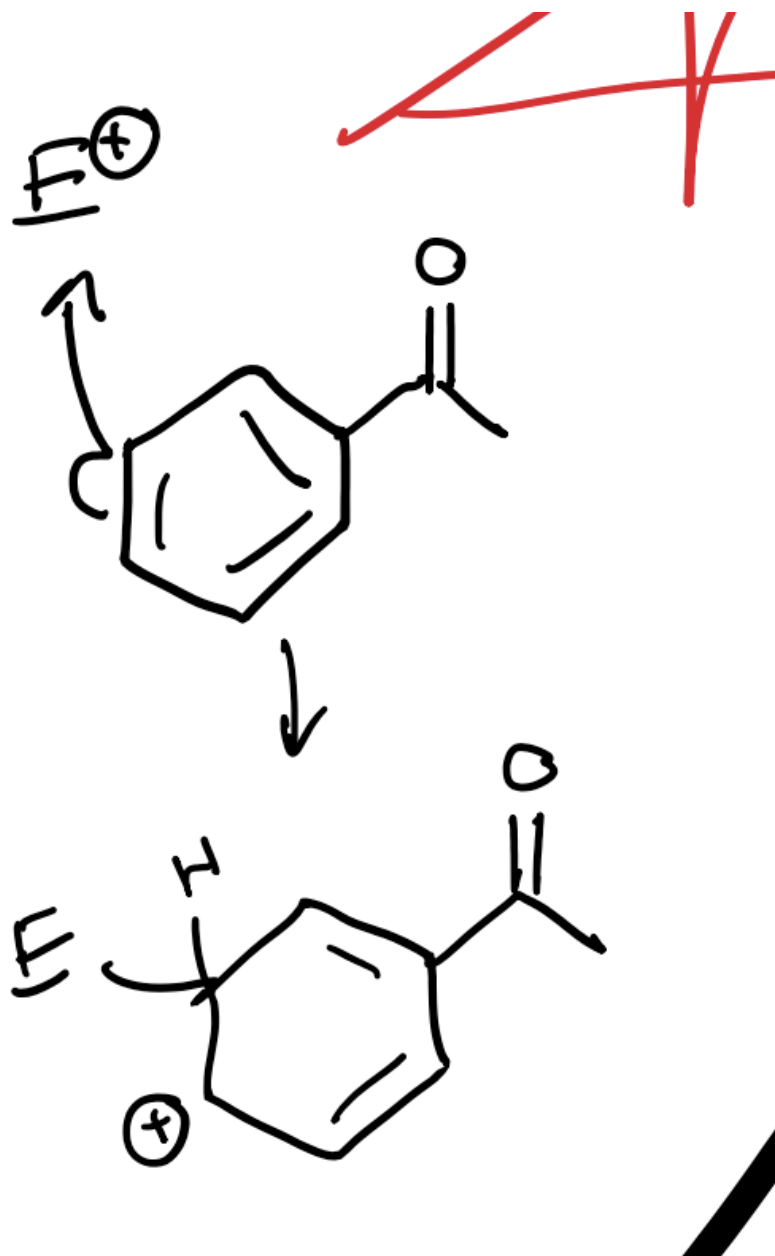


Figure 50



Figure 51: caption

Core Concept 1.18: Halogens as Directing Groups

Halogens can also act as directing groups, and based on data, they act as O/P directors. To understand the underlying basis of this observation, let's analyze monohalobenzene in the context of the inductive effect and resonance.

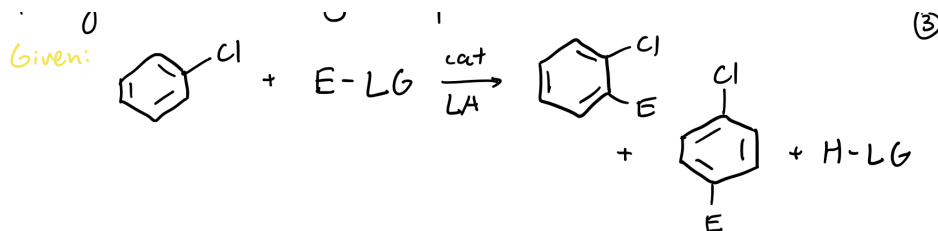


Figure 52

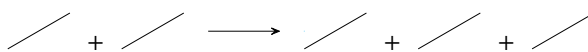


Figure 53: caption

(a) Inductive Effect

Figure 54



Figure 55: caption

Under an O/P directing EAS interaction, inductive analysis would actually discourage O/P directing because the subsequent carbocation transition state places a carbocation that already has lowered e^- density even lower by residing next to the highly electronegative halogen atom. As a result, the dipole moment created destabilizes the structure.

(b) Resonance

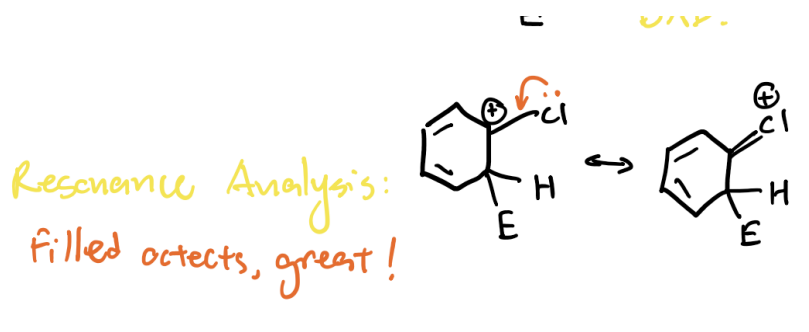


Figure 56

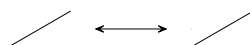


Figure 57: caption

Resonance, on the other hand, allows the halogen to donate e^- density in order to fill all octet requirements throughout the entire molecule, which favors the O/P directing EAS interaction.

Although these two effects contradict one another, we can confidently say that the effects of resonance trump the inductive effect because real data in practice results in the O/P directing as favored.

1.2.3 Aromatic Substituent Manipulations

Core Concept 1.19: Reduction of Acyl Ketone

The reduction of an acyl ketone is a reduction reaction that was previously used in Core Concept 1.11 to replace a carbonyl oxygen with two hydrogen bonds.

⑥ Reduction of Acyl ketone

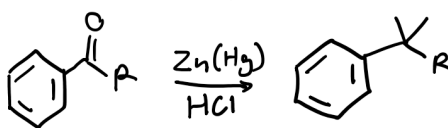


Figure 58



Figure 59: caption

No mechanism is provided for this reaction.

Core Concept 1.20: Nitrobenzene Reduction to Aniline

The reduction of nitrobenzene work similarly to the reduction of acyl ketone (Core Concept 1.19) wherein oxygens are replaced by hydrogens. This reaction uses the same reactants to carry this replacement out.

② Nitro Reduction

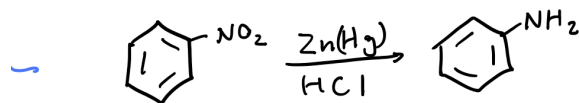


Figure 60

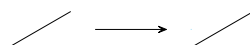


Figure 61: caption

No mechanism is provided for this reaction.

Core Concept 1.21: Diazotization of Aniline

The diazotization of aniline is a potential subsequent reaction to nitrobenzene reduction as the product of the latter is the reactant of the former.

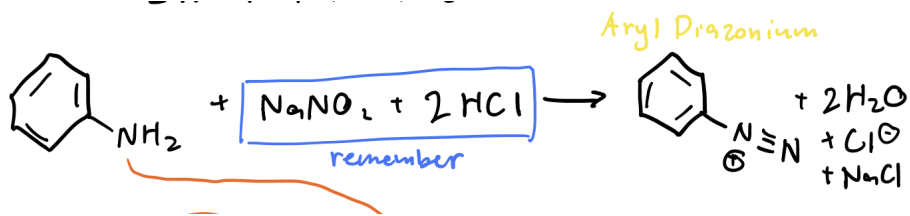


Figure 62

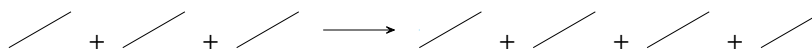


Figure 63: caption

The reaction mechanism occurs as follows, first starting with the conversion of NO_2^- to a nitrosyl cation.

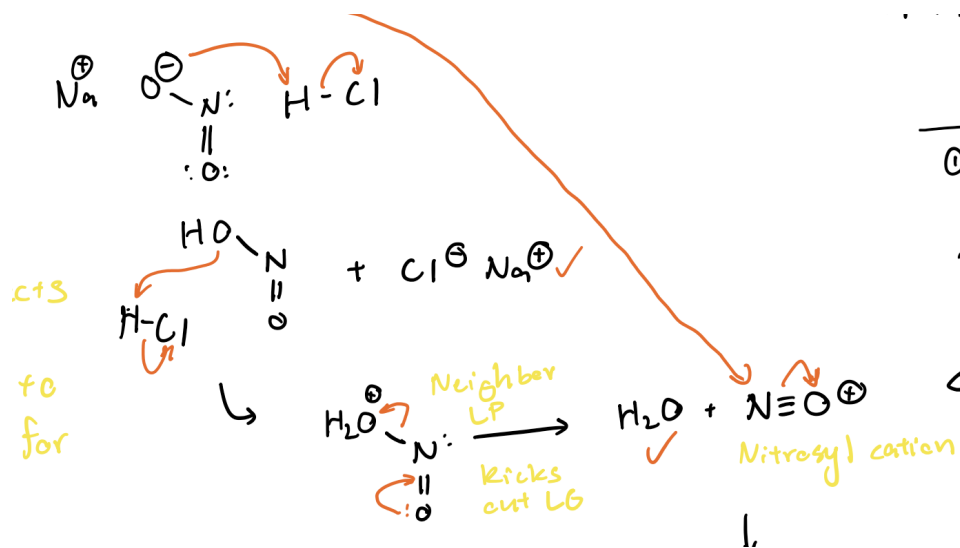


Figure 64

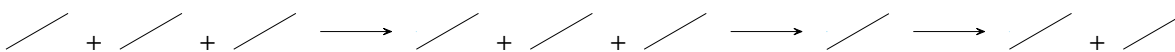


Figure 65: caption

The highly reactive nitrosonium cation then goes onto interact with the aniline in the following manner to produce **aryl diazonium**.

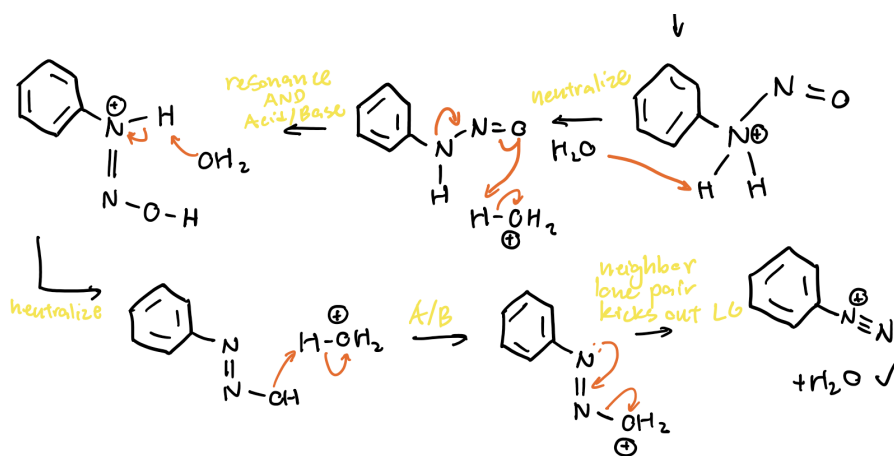


Figure 66



Figure 67: caption

1.3 Nucleophilic Aromatic Substitution

Core Concept 1.22: Nucleophilic Aromatic Substitution

Nucleophilic aromatic substitution (NAS or S_NAr) is an interaction in which the aromatic ring is the electrophile. NAS comes in two "flavors": **diazonium** (Core Concept 1.23) or **addition/elimination** (Core Concept 1.24) chemistry.

Recall that in electrophilic aromatic substitution (Subsection 1.2), the aromatic ring is the nucleophile.

Core Concept 1.23: Diazonium Chemistry

Diazonium chemistry begins by N_2 leaving a phenyldiazonium molecule—via breaking the bond between the σ $C(sp^2)-N(sp)$ bond—to produce a carbocation possessing benzene ring and an extremely stable N_2 gas molecule.

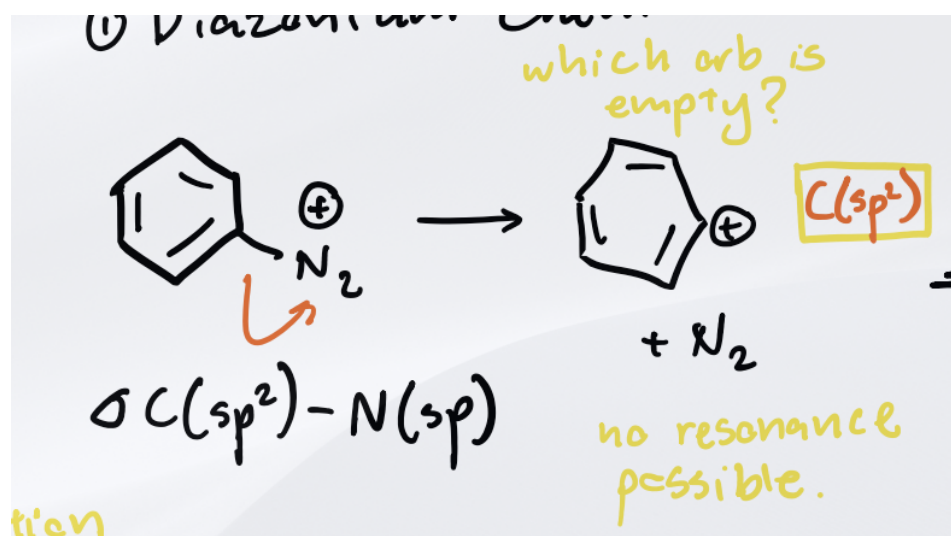


Figure 68

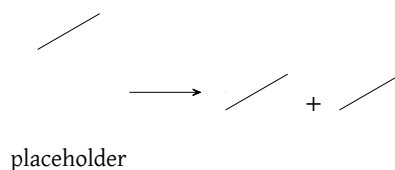


Figure 69: caption

It's important to notice that resonance is not possible in this specific carbocation state because the orbital that is empty is the $C(sp^2)$ orbital. We would need an unbalanced p orbital for resonance to occur. As a result, this phenyl cation is highly reactive and hungry for any nearby nucleophile to help stabilize its positive charge.

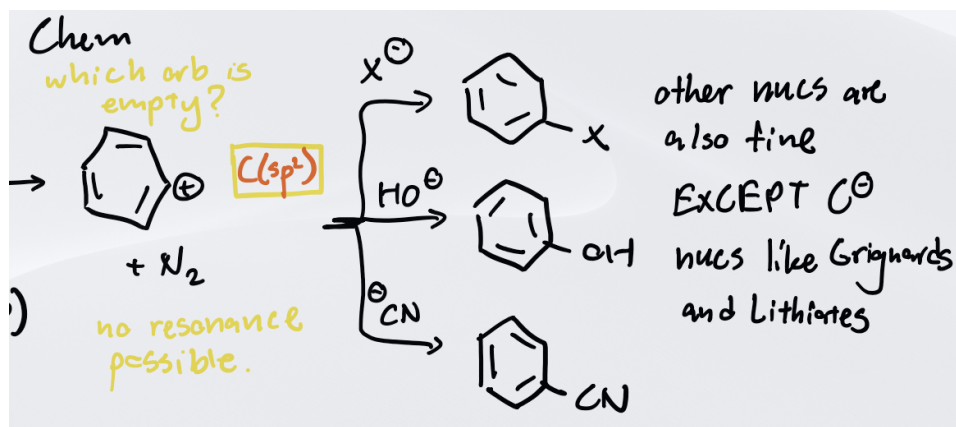


Figure 70

figureouthowtodomultiplearrows

Figure 71: caption

As shown above, nearly all covered nucleophiles work in this interaction **with the exception of grignards and lithiates**. The reasoning behind this will not be covered in the scope of these notes.

Combining this diazonium chemistry with previously learned reactions, chemists can effectively attach nearly any nucleophile to a benzene molecule using the following reaction pathway.

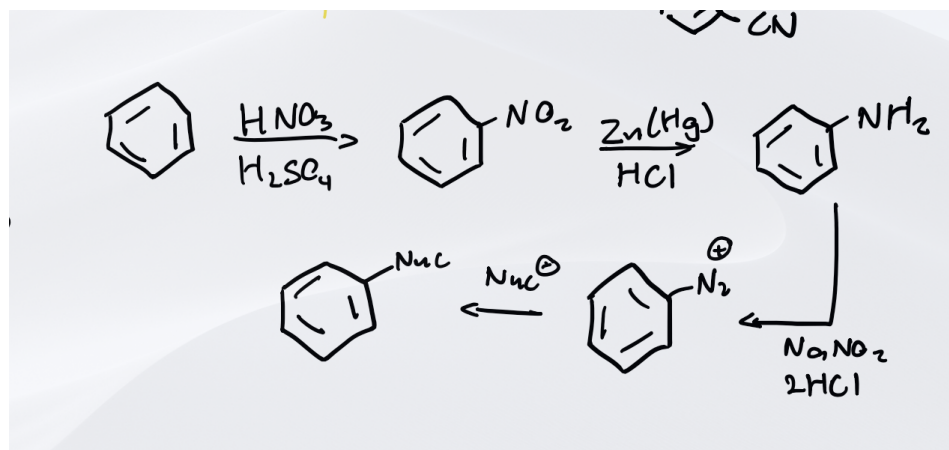


Figure 72: reference the core concepts for the reactions within the caption



Figure 73: caption

Core Concept 1.24: Addition/Elimination

The NAS **addition/elimination** reaction occurs in the following general manner.

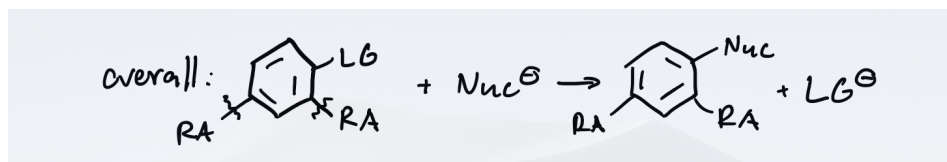


Figure 74

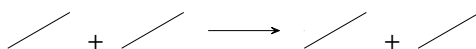


Figure 75: caption

Recall that we have seen resonance acceptors in resonance M directors (*Core Concept 1.17*), however, these resonance acceptors are not for the purpose of directing meta in the context of NAS chemistry. Their main purpose is to be able to accept e^- density after the nucleophile attacks the carbon with the leaving group.

Additionally, this reaction requires strong nucleophiles because it ultimately needs to break aromaticity as shown in the mechanism below. It should be noted that this step—the addition—is consequently the rate-determining step.

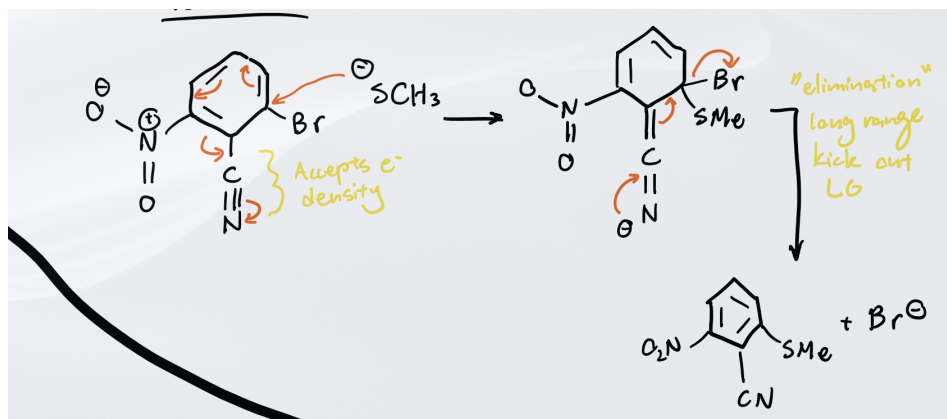


Figure 76



Figure 77: caption

As the mechanism showcases, the reaction is simply a relay of e^- density to the resonance acceptor, which then bounces back the e^- density in order to kick out the leaving group in the end. However, since the outcome of this interaction is ultimately the same as that of an S_N2 reaction, one may wonder why S_N2 is not a viable pathway.

Revisiting the S_N2 mechanism, recall that, in the lense of molecular orbitals, the nucleophile interacts with the σ^* anti-bonding orbital between the electrophile and leaving group in order to kick out the leaving group and create a bond simultaneously via a backside attack.

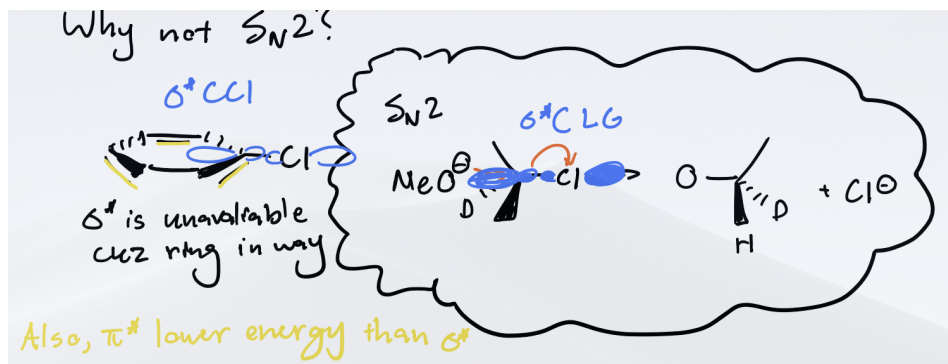


Figure 78

Figure 79: caption

In the aromatic ring scenario, this necessary σ^* interaction is extremely unlikely due to the physical barrier to entry because of the ring itself.

In the context of the frontier molecular orbital theory, the π^* orbital is lower in energy than σ^* , making it the LUMO and a much more accessible orbital to interact with, which is what happens in addition/elimination, the favorable pathway.

Lets go through two examples.

(a) **Example 1: Predicting Major Kinetic Product**

Consider the following reaction consisting to two possible leaving groups F and Cl with a NO_2 resonance acceptor.

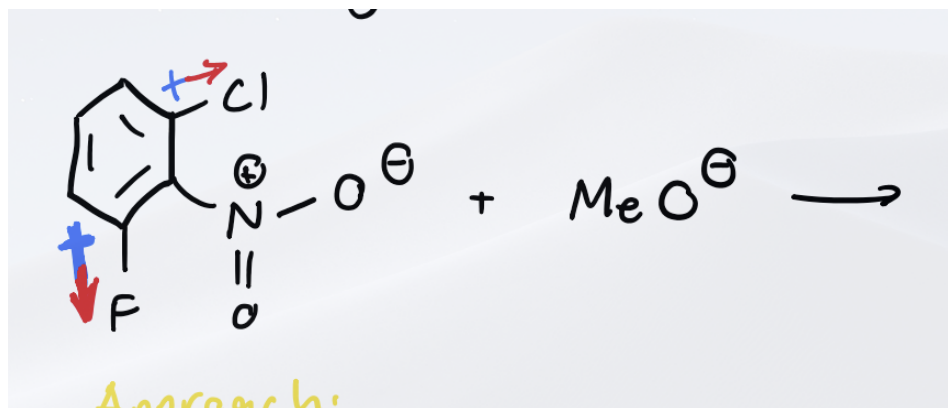


Figure 80

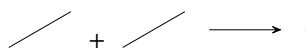


Figure 81: caption

Both addition/elimination pathways are possible as shown below, ultimately leading to two distinct products.

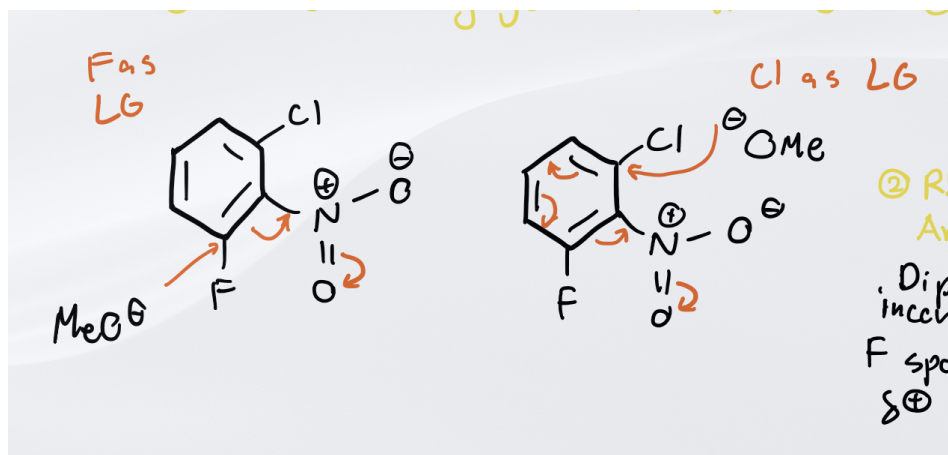


Figure 82



Figure 83: caption

Since the nucleophile is held constant across both reaction pathways, the ability of the nucleophile to break aromaticity is not a factor. Instead, it is the greater dipole moment created by the more electronegative fluorine over the chlorine. Due to the greater partial positive charge of the fluorine carbon, MeO^- favors attacking that carbon because it is more electrophilic. Thus the reaction favors the substitution of the F over the Cl.

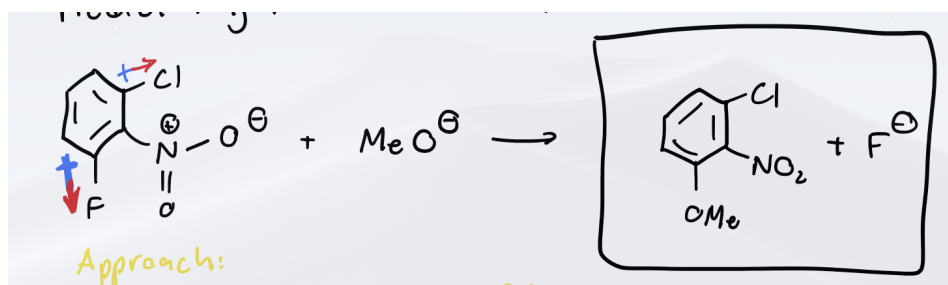


Figure 84



Figure 85: caption

(b) Example 2: Predicting Product of NAS Chemistry

Consider the following addition/elimination reaction below.

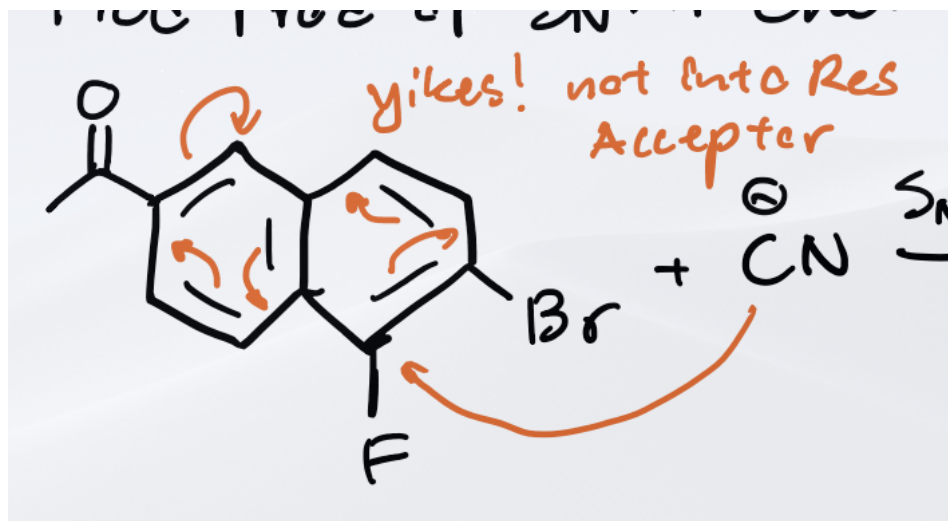


Figure 86: short initial reactants without arrows



Figure 87: caption

In this case, simply only one of the leaving group substitutions work to donate e^- density into the resonance acceptor and the other doesn't.

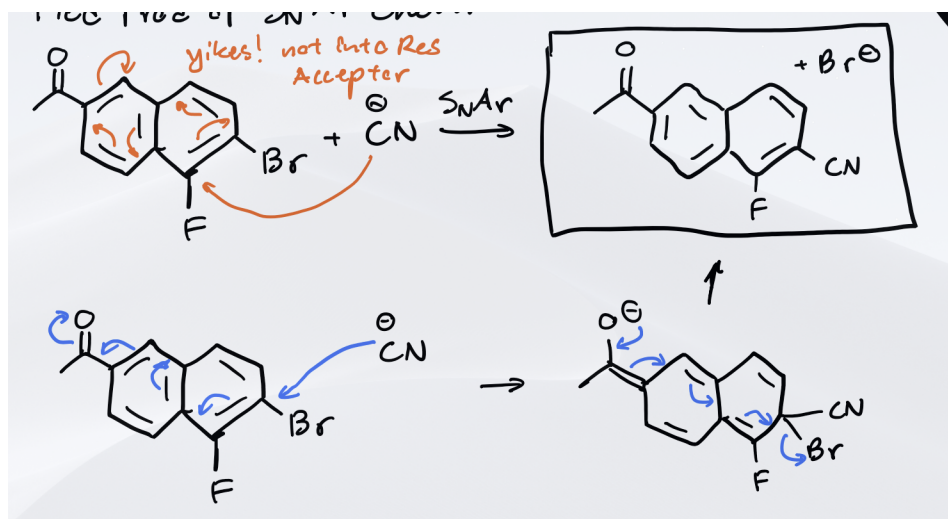


Figure 88: showcase the pathways and how one isn't possible

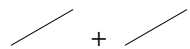
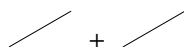


Figure 89: caption

Thus the exclusive reaction outcome maps out as follows.

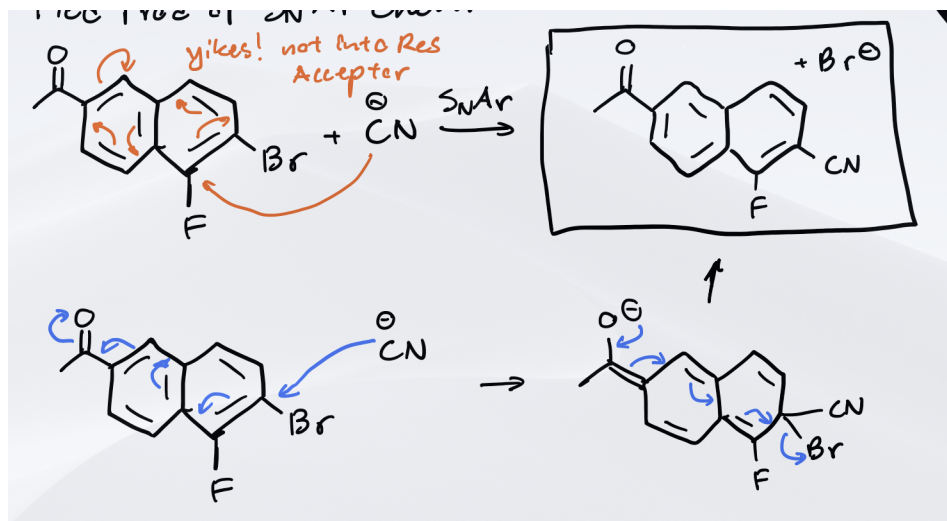


Figure 90: complete chemical equation

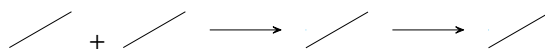


Figure 91: caption

(c) Example 3: Predicting Starting Material

Consider having to determine the starting aromatic compound to the following products based on the given nucleophile.

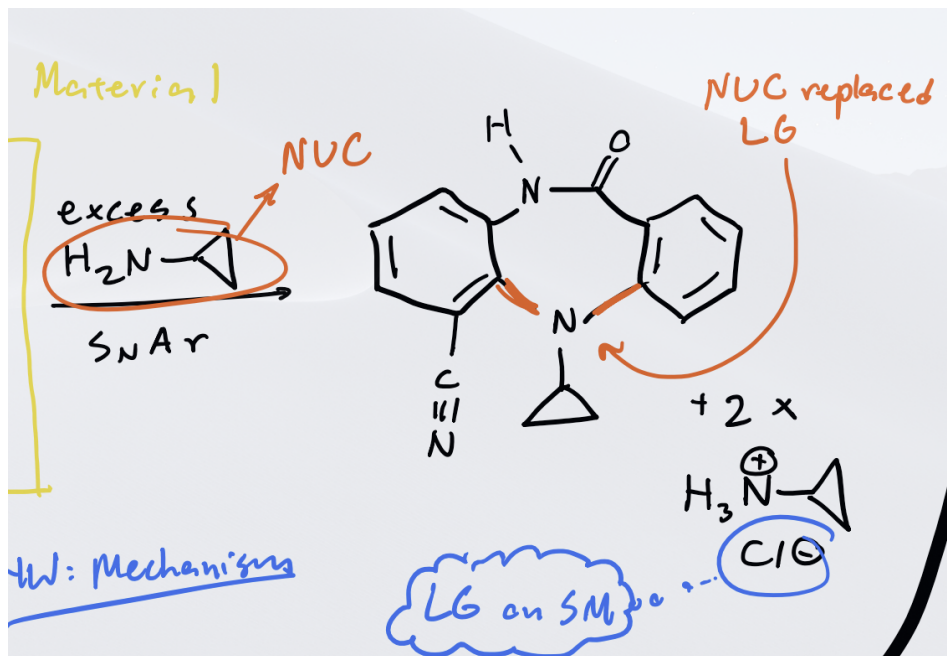


Figure 92

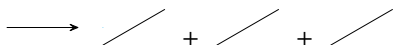


Figure 93: caption

Because we see two Cl^- and two $\text{H}_3\text{N}^+\text{R}$ on the product side, we know that the addition/elimination has occurred twice. Luckily the product quite literally has the nucleophile placed directly on center connected to the aromatic ring system via two bonds on the nitrogen. Based on this, we can reasonably predict the starting material to have had these two bond substituted.

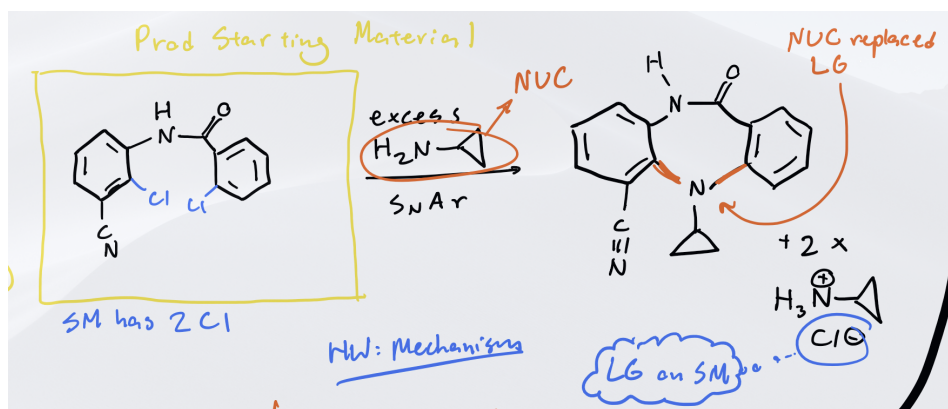


Figure 94



Figure 95: caption